

Detailed HTML Notes (Complete Beginner to Forms)

Prepared based on your syllabus PDF

1. What is HTML & WWW

- WWW is a network of linked web pages accessed using browsers.
- HTML is the language used to structure content on the web.
- HTML files are plain text with tags that browsers understand.
- Invented by Tim Berners-Lee (1991) to share scientific documents.
- HTML5 is the modern version used today.

2. Basic Structure of an HTML Page

- tells the browser we are using HTML5.
- is the root element.
- contains title and metadata (not visible).
- appears on browser tab.
- contains visible content.

3. Elements, Tags & Attributes

- Elements are components like headings, paragraphs, images, tables.
- Tags define elements: opening and closing .
- Standalone tags have no closing tag (br, img).
- Attributes add extra information inside opening tag.
- Common attributes: id, class, src, href, style.
- Attribute values must be written inside quotes.

4. Text Formatting Tags

- Headings: h1 to h6 show content hierarchy.
- Paragraph: p separates text into blocks.
- Line break: br forces next line.
- Bold: b, Italic: i.
- Pre: preserves spacing and uses monospace font.
- Comments: hidden from browser.

5. Links (Anchor Tag)

- a tag creates hyperlinks.
- href specifies destination.
- target='_blank' opens in new tab.
- mailto creates email links.
- Links can be relative (same folder) or absolute (full URL).

- Named anchors allow jumping to sections inside a page.

6. Images

- `img` tag embeds pictures.
- `src` attribute specifies image path.
- `alt` text appears if image fails to load.
- `width` and `height` control image size.
- Relative paths point inside your project folder.
- Absolute paths point to external websites.

7. Lists

- `ul` creates bullet list.
- `ol` creates numbered list.
- `li` defines list item.
- `ol` type attribute changes numbering style (A, a, I, i).

8. Tables

- `table` defines table structure.
- `tr` = row, `th` = header cell, `td` = data cell.
- `border` shows table lines.
- `cellpadding` adds space inside cells.
- `cellspacing` adds space between cells.

9. Block vs Inline Elements

- Block elements start on a new line (`p`, `div`, `ul`, `h1`).
- Inline elements stay in the same line (`a`, `span`, `img`).
- `div` groups large sections for styling.
- `span` styles small parts of text.

10. HTML Entities

- `<` shows `<` symbol.
- `>` shows `>` symbol.
- `&` shows `&` symbol.
- `"` shows double quotes.
- inserts fixed space.

11. Forms & User Input

- `form` collects data from users.
- `action` defines where data is sent.
- `method` can be GET (URL) or POST (secure).
- input types: `text`, `password`, `radio`, `checkbox`, `hidden`, `submit`.

- textarea allows multi-line input.
- select & option create dropdown menu.
- Radio buttons with same name form a group.
- Checkbox allows multiple selections.

12. HTML5 Semantic Elements

- header defines top area.
- nav contains navigation links.
- section groups related content.
- article represents independent block.
- main is central content.
- footer defines bottom info.